A Coarse-to-Fine Scene Matching Method for High-Resolution Multiview SAR Images

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***Abstract*— Scene matching involves establishing correspondences between multiple images of the same location and poses significant challenges for synthetic aperture radar (SAR) images due to the anisotropic scattering prosperities of SAR targets; variations in looking and azimuth angles further complicate the matching process. A matching algorithm is proposed based on a coarse-to-fine framework to address these issues. First, a coarse matching employing normalized cross correlation (NCC) with a sliding window is applied to filter out irrelevant regions, reducing distractions, and shortening the processing time. Subsequently, a Siamese neural network (SNN), incorporating ResNet-50 and convolutional block attention module (CBAM) for enhanced feature extraction, is introduced to learn and discern differences between inputs. The effectiveness and robustness of the proposed method are validated through extensive experiments using a self-made dataset derived from Umbra Satellite.**

***Index Terms*— Anisotropy, coarse-to-fine, deep learning, scene matching, synthetic aperture radar (SAR).**

I. Introduction

SCENE matching is used to relate the same or similar contents from two or more images and has been widely applied to 3-D reconstruction [1], image fusion [2], and image retrieval [3].

There are two major issues in scene matching generally: what is used for matching and how to perform the matching [4]. A direct method known as template matching compares two images by the original image pixel or information instead of detecting any image structures. Common algorithms in this category include mean absolute differences (MADs) [5], sequential similarity detection algorithms (SSDAs) [6], normalized cross correlation (NCC) [7], and so on, which exploit grayscale information at a pixel level primarily, and originally developed for medical imaging. Frequency-domain information can also, furthermore, be used to suppress noise and interference [8], [9], [10]. One drawback of template matching, however, is that it has notable limitations that are computationally intensive and sensitive to nonlinear deformation.

Another category called the feature-based method is developed to address the limitations of template matching by focusing on points of interest in images. Harris and Stephens [11], SIFT [12], and SURF [13] are typical traditional methods, and with their robustness to transformations like rotation and scaling, widely used in image scene matching. Rublee et al. [14] introduced Oriented FAST and Rotated BRIEF (ORB) which allows the acquiring of a descriptor efficiently and robustly. Additionally, due to the ability to deep feature extraction and resilience in nonlinear variations, deep learning has become a mainstream direction. Convolutional neural networks (CNNs) such as AlexNet [15], VGG [16], and MobileNets [17] are the core algorithm models for feature extraction and description owing to their achievements in the ImageNet Large Scale Visual Recognition Challenge. LIFT for similarity measurement proposed by Yi et al. [18] is the first end-to-end neural network to effectively realize all matching steps at the same time. A self-supervised network named SuperPoint [19] can extract pixel-level features, and SuperGlue [20], developed by the same team, reformulates the matching problem as a differentiable optimal transfer solution problem, both of which have achieved remarkable results. Overall, numerous experiments have demonstrated that deep learning shows significantly superior performance over traditional methods and presents promising potential.

Naturally, these algorithms were initially applied to optical images, given their wide availability. Compared to optical images, synthetic aperture radar (SAR) images, however, have the advantages of capturing topographical features with greater precision and have a higher resolution under extreme meteorological conditions, and SAR has proven to be an effective tool for target detection [21], navigation [22], and so on. As a result, the matching algorithms mentioned above have been applied to the SAR scene matching field. In the case of template matching, they were first used in remote sensing in 2000 [23] but were limited by their arithmetic speed and accuracy. Feature-based methods, widely adopted in remote sensing, have achieved more success.

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SIFT and others, however, struggled to give a satisfactory performance in remote sensing image matching due to the fundamentally different imaging principles involved. Some researchers modified these descriptors for SAR images and obtained SAR-SIFT [24], USAR-SIFT [25], and I-SAR-SIFT [26] to be more robust to coherent noise while the feature points are evenly distributed by an innovative approach to enhance uniqueness. Deep learning algorithms for SAR scene matching have also been introduced to the field. Serverira et al. [27] put forward an algorithm for Arctic SAR images, which processed input feature points through neural networks. Wang et al. [28] proposed a network to realize registration, tackling the challenge of insufficient SAR images, and offered a new deep-learning idea for scene matching in the realm of remote sensing multimodes. Merkle et al. [29] first applied a Siamese neural network (SNN) for optical and SAR images using dilated convolution to expand the receptive field. He et al. [30] conducted remote sensing image matching in complex backgrounds by a Siamese CNN.

Although many previous works on SAR scene matching have focused on dense matching techniques, such as image registration and alignment, there has been a recent shift toward patch-based matching that compares with stored images or panoramic images to identify the most similar ones. Semenov et al. [31] performed such a task effectively with SNN and SqueezeNet measuring similarity. Their method, however, is limited to images captured from identical looking and azimuth angles, failing to account for the influence of varying angles (as shown in Fig. 1), which renders it unsuitable for practical matching tasks. In this article, we address this limitation by incorporating both looking and azimuth angles. We use Umbra Satellite SAR images taken from different looking and azimuth angles as a dataset. A two-step matching process is adopted: firstly, coarse matching filters out irrelevant but similar regions caused by angle variations and scattering to enhance accuracy and reduce processing time; secondly, the Siamese model in fine matching extracts deeper representative features, which helps mitigate the impact of noise and angular differences, thus facilitating automatic similarity matching. Once the correct image region is identified, the precise location is marked. The effectiveness of the proposed coarse-to-fine algorithm is validated through a series of matching experiments. Our method provides real-time identification of potential matching regions, making it promising for applications such as SAR-based navigation systems and aircraft trajectory correction.

This article is organized as follows. Section II provides basics about the structure of ResNet and SNN, and the self-made dataset is explained as well. The proposed algorithm for efficient and accurate scene matching is presented in Section III. Experiments and analysis are presented in Section IV, and the conclusion is in Section V.

II. Related Work

A. ResNet

As an efficient and classical CNN for image process tasks, ResNet is often used as the backbone. ResNet takes a large number of residual blocks to face the degradation matter. Relative experiments suggest that ResNet can extract deeper and more effective features to improve model performance on public optical datasets. Fig. 2 illustrates the two residual blocks used in ResNet.

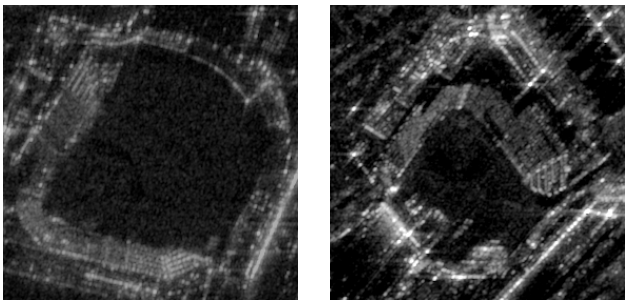


Fig. 1. Two SAR images were taken at different times in Fort Wayne; due to differences in satellite position, the looking angles and azimuth angles are different, leading to different features.

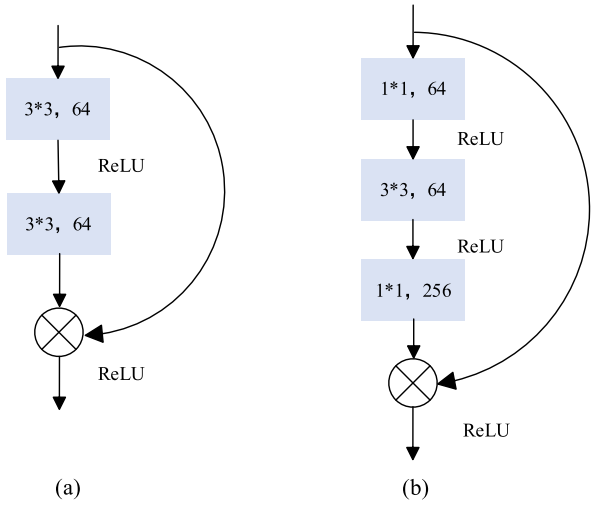


Fig. 2. Two residual blocks of ResNet [32]. (a) Residual block 1. (b) Residual block 2.

For Fig. 2(a), it applies to ResNet-18/34, and there are two 3×3 convolutional layers. Assuming there is an input, the block leads to

$$y = x + F(x), \quad x_{\text{out}} = r(y) \tag{1}$$

where x_{in} and x_{out} represent inputs and outputs of the residual unit, respectively, and $F(\cdot)$ denotes the functions that residuals learned by the network. x denotes the constant transform, r means the ReLU function.

For Fig. 2(b), it applies to ResNet-50/101/152. The two convolutional layers with a specific filter size of 3×3 are replaced by three convolutional layers: $1 \times 1 + 3 \times 3 + 1 \times 1$, allowing the dimensionality of the input to be adjusted accordingly. This approach eliminates the need to modify the number of kernels in the convolutional layers to fit the input data, ensuring that the outputs do not affect the kernel settings of the subsequent layers. The final layer subsequently restores the input dimension. This transformation significantly reduces computational complexity while preserving the integrity of the extracted information, thereby maintaining the model's accuracy.

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TABLE I
STRUCTURAL PARAMETERS OF SOME RESNET NETWORKS

Layer name	Output	34 layers	50 layers	101 layers
conv	112×112	7×7 , 64, stride=2		
		3×3 , stride=2		
layer 1	56×56	$\begin{bmatrix}3\times3,64\\3\times3,64\end{bmatrix}\times3$	$\begin{bmatrix}1\times1,64\\3\times3,64\\1\times1,256\end{bmatrix}\times3$	$\begin{bmatrix}1\times1,64\\3\times3,64\\1\times1,256\end{bmatrix}\times3$
layer 2	28×28	$\begin{bmatrix}3\times3,128\\3\times3,128\end{bmatrix}\times4$	$\begin{bmatrix}1\times1,128\\3\times3,128\\1\times1,512\end{bmatrix}\times4$	$\begin{bmatrix}1\times1,128\\3\times3,128\\1\times1,512\end{bmatrix}\times4$
layer 3	14×14	$\begin{bmatrix}3\times3,256\\3\times3,256\end{bmatrix}\times6$	$\begin{bmatrix}1\times1,256\\3\times3,256\\1\times1,1024\end{bmatrix}\times6$	$\begin{bmatrix}1\times1,256\\3\times3,256\\1\times1,1024\end{bmatrix}\times23$
layer 4	7×7	$\begin{bmatrix}3\times3,512\\3\times3,512\end{bmatrix}\times3$	$\begin{bmatrix}1\times1,512\\3\times3,512\\1\times1,2048\end{bmatrix}\times3$	$\begin{bmatrix}1\times1,512\\3\times3,512\\1\times1,2048\end{bmatrix}\times3$
	1×1	average pooling, 1000-d FC, softmax		

The construction of ResNet mainly refers to the structure of VGG, adhering to two basic principles: if the output feature size of two layers remains the same, it implies that these layers should possess an equal number of convolutional kernels. Conversely, if the output feature map size is halved, then double kernels are used to maintain the dimensionality. Its output comprises a global average pooling and a fully connected layer and outputs a 1000-D vector via a softmax classifier. ResNet variants can be divided into five stages, with the structural parameters of specific networks in Table I.

B. Siamese Neural Network

SNN [33] is an architecture designed for learning similarity between paired samples. Its primary objective is to learn a mapping that embeds the input data into the metric space, where the distance between the embedded samples is representative of their "semantic" similarity. The ultimate goal is to correlate the output with the input pairs, seeking the optimal system parameters that minimize the output. Fig. 3 illustrates its framework. The input consists of an image pair, and the corresponding label L is binary. If the inputs belong to the same category, then $L = 0$; otherwise, $L = 1$. W is the parameter that the network will share and learn. $f_W(\text{Int}_1)$ and $f_W(\text{Int}_2)$ represent the low-dimensional embeddings generated by the feature extraction network with shared weights. These embeddings are typically computed without classifiers, using the backbone of a classification model, such as VGG. The shared weights can be achieved by constructing two identical subnetworks. The similarity between the inputs is then quantified by a distance measure, which is defined by a similarity metric function as given as follows:

$$D_W = F(f_W(\text{Int}_1) - f_W(\text{Int}_2)). \tag{2}$$

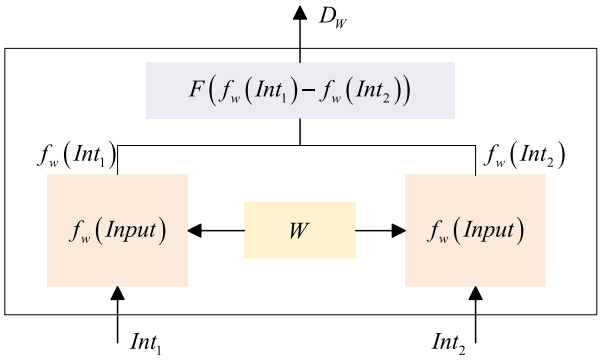


Fig. 3. Structure of the SNN.

Commonly used similarity metric functions $F(\cdot)$ in SNN include $L1$ norm, Euclidean distance, and cosine distance, which are defined separately in [34].

Loss functions for SNN include binary cross entropy (BCE) loss with sigmoid, contrastive loss, as well as triplet loss.

The BCE loss function is mainly aimed at binary classification tasks. Given that the result of scene matching can be categorized as either "similar" or "dissimilar," it effectively frames the problem as a binary classification task, and its definition is given by

$$\text{BCE} = -\frac{1}{\text{SUM}}\sum_{i=1}^{\text{SUM}}\left[l_i \cdot \log(f_{\text{out}}) + (1 - l_i) \cdot \log(1 - f_{\text{out}})\right]. \tag{3}$$

One basic criterion for contrastive loss is that the distance between similar inputs should be small, while that for dissimilar inputs should be large. It is defined as follows:

$$\text{Cont} = \frac{1}{2\text{SUM}}\sum_{i=1}^{\text{SUM}}\left(t_i D_i^2 + (1 - t_i) \cdot \max(0, \text{margin} - D_i)^2\right). \tag{4}$$

The triple loss trains the model by triads. There is one positive sample, one negative sample, and one anchor sample, and the aim is to make the anchor sample closer to the positive one and farther away from the negative one [22]. The graph is given in Fig. 4

$$\text{TriLoss} = \max(d(a, p) - d(a, n) + \text{margin}, 0). \tag{5}$$

C. Dataset

Since deep-learning-based SAR scene matching has been scarcely explored, obtaining SAR image matching datasets is challenging. Our proposed model is trained and evaluated on a self-made dataset, which consists of SAR images captured by the Umbra Satellite in 2023. This satellite provides the highest resolution SAR images commercially available, with a resolution better than 25 cm (10 in). Additionally, the Open Data Initiative monitors ten distinct locations worldwide, offering a database of SAR panoramic images with spatial resolutions of 0.5 and 1 m. Representative images from the dataset are shown in Fig. 5.

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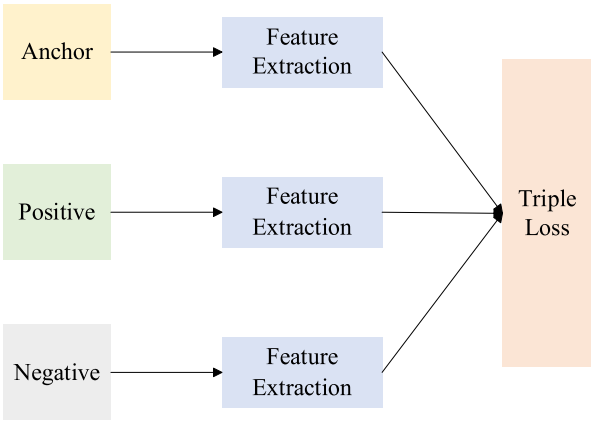


Fig. 4. Triple Loss.

SAR images of Fort Wayne taken on June 14, 18, and 19, 2023, respectively, are in Fig. 5(a)–(c), with a spatial resolution of 0.5 m and a pixel dimension of 6281 × 6280; Fig. 5(d)–(f) are taken from Johnston, Iowa Farm on July 23, 25, and 26, 2023, with the same spatial resolution, but a different pixel dimension of 6665 × 6665; Fig. 5(g)–(i) are of North Dakota State University Plot with a 1 m spatial resolution and a pixel dimension of 6605 × 6605 taken on July 23, 24, and 25, 2023, respectively.

Slices of landmarks with more significant features are cropped from the panoramas and scaled without distortion to a size of 280 × 280. Since the initial image slices obtained are insufficient for the training of the complex model, a data expansion strategy is adopted. The cropped images are transformed by flipping, rotating, scaling, and brightness comparison to obtain groups of images. Part of the dataset is shown in Fig. 6. Images at the same location, different resolutions, different periods, and different angles are selected as positive samples (which are similar), and negative sample relationships are formed between samples from different locations (which are dissimilar). The final dataset contains 3185 images of 37 categories, divided into training set, validation set, and test set with a ratio of 8:1:1.

III. Methodology

Considering the anisotropic scattering targets and speckle noise inherent in SAR images, along with the limited availability of experimental data, we adopt a coarse-to-fine algorithmic framework for our study. The entire framework is illustrated in Fig. 7. In the coarse matching stage, we employ template matching with a sliding window to isolate several relevant regions from the panorama, thereby reducing both the search space and computational cost. For fine matching, we use a modified SNN based on ResNet-50, incorporating an attention module to obtain deeper features to avoid distractions from irrelevant information, enhancing both accuracy and efficiency. SAR images from the Umbra satellite, captured with varying looking and azimuth angles, are used to enhance the capability for the model's generalization.

A. Template Matching Method for Coarse Matching

On the one hand, the anisotropic nature of SAR images can cause features extracted from different regions to appear quite similar. Coarse matching can help isolate the most relevant region while filtering out irrelevant interference. Additionally, in [31], the matching process is based on image patches, which requires deep learning models to process the data block by block. This approach, however, is time-consuming. In order to mitigate the problem, a coarse matching step is necessary to identify the relevant region and reduce the search area for subsequent fine matching.

While both template matching and hand-crafted feature descriptors can perform matching through correlation or feature similarity comparison, the latter is particularly vulnerable to speckle noise and background interference that are common in SAR images. This susceptibility often results in inaccurate screening due to angular discrepancies. In contrast, template matching uses grayscale information directly within a localized region. By limiting the search area, it reduces the influence of visually similar regions, leading to more accurate results; thus, this article adopts template matching for the preliminary process.

The template matching process is shown in Fig. 8. In this process, we begin with a panoramic image and a query image. A reference coordinate system is established on the panoramic image, from which a subimage is extracted. The correlation between the query image and the extracted subimage is then computed. This results in a similarity matrix, where each element represents the correlation between various locations on the panoramic image and the query image. The matching area can subsequently be determined based on the values of the elements in this matrix.

The key to ensuring the effectiveness of template matching lies in the similarity metric function. We choose NCC as the similarity metric function here [35], [36]. In NCC, the range of function value is [0, 1], with closer to 1 meaning a stronger correlation and closer to 0 indicating a worse correlation.

When NCC is applied to an image, the image can be viewed as a 2-D dataset, and the pixel values representing each point can be viewed as an array of grayscales, so the correlation coefficient for a 2-D image can be obtained [7]

$$\mathrm{NCC} = \frac{\sum_{x_2,y_2}\left(Q'(x_2,y_2)\cdot S'(x_1+x_2,y_1+y_2)\right)}{\sqrt{\sum_{x_2,y_2}Q'(x_2,y_2)^2\cdot\sum_{x_2,y_2}S'(x_1+x_2,y_1+y_2)^2}} \tag{6}$$

where $*'(x,y) = *(x,y) - 1/(mn)\sum_{x,y} *(x,y)$.

In order to address the challenges encountered when applying this method to SAR images, such as unclear peak points and potential omissions at window boundaries, we adopt a specialized sliding window approach for NCC computation.

The process is depicted in Fig. 9. A sliding window is applied to the panoramic image, with the query image serving as the template for computing NCC. After the NCC is computed, a matrix is generated. The elements of this matrix are then sorted, and the maximum NCC value, along with its corresponding window area, is recorded. Once the NCC calculations for all windows are completed, the maximum values that exceed a specified threshold (which can be adjusted as needed) are identified. Finally, the window areas

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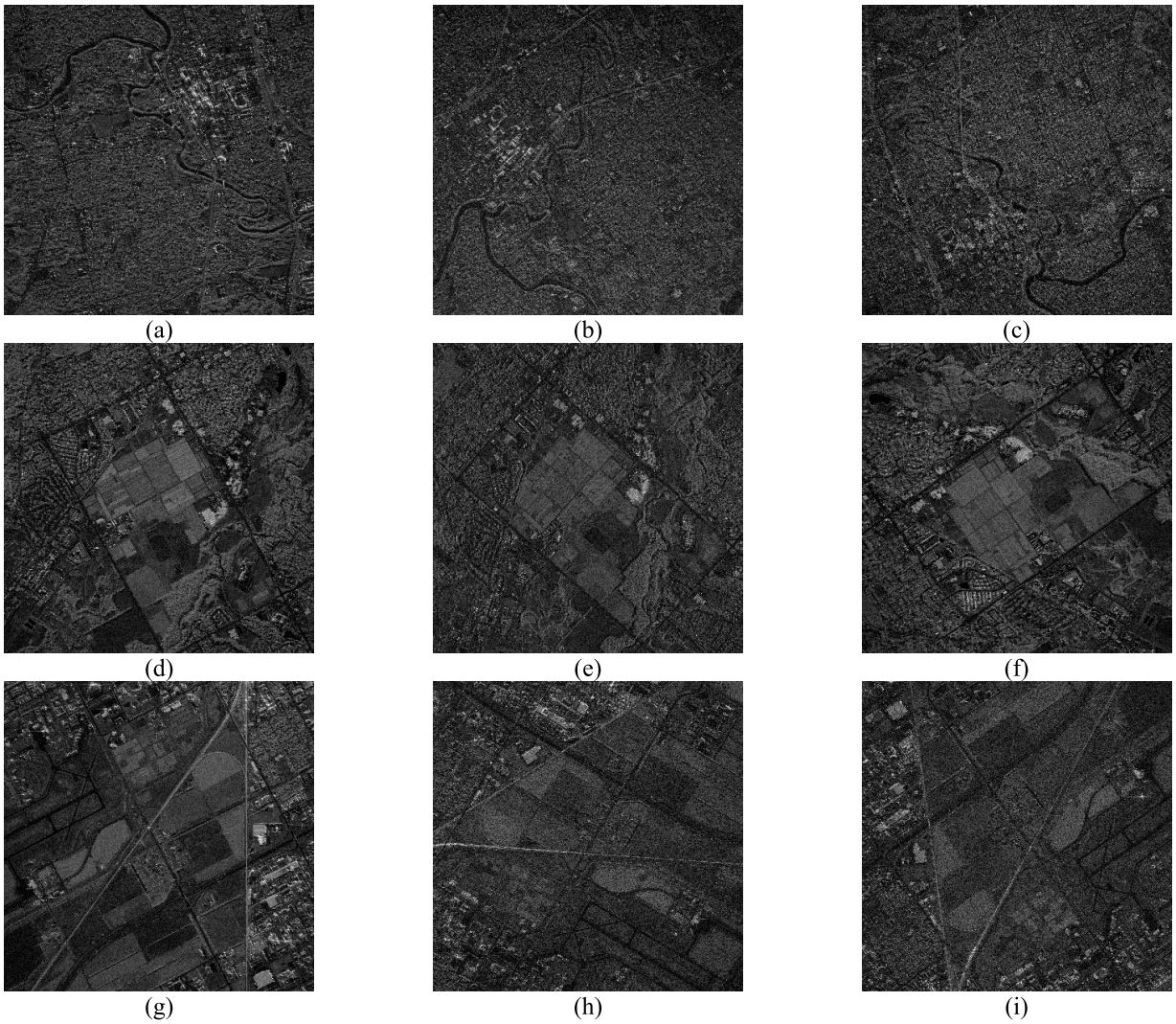


Fig. 5. Several available SAR images from Umbra Satellite taken in 2023. (a) Fort Wayne, June 14. (b) Fort Wayne, June 18. (c) Fort Wayne, June 19. (d) Johnston, Iowa Farm, July 23. (e) Johnston, Iowa Farm, July 25. (f) Johnston, Iowa Farm, July 26. (g) North Dakota State University, July 23. (h) North Dakota State University, July 24. (i) North Dakota State University, July 25.

corresponding to the selected elements are extracted, yielding the coarse matching results.

B. Attention-Based Backbone: ResNet-50 With CBAM

During fine matching, the input images are processed by CNN to get their feature maps first. Taking ResNet as the backbone for SAR images is also a workable method. As we can see in Table I, the depth of each ResNet version varies, and the model accuracy increases as the depth increases. The increase in network depth, however, also makes the network more complex with large parameters and slower convergence speed. Considering the tradeoff between effectiveness and complexity, we choose ResNet-50 as the backbone of the feature extraction part without a classifier.

After being processed by convolutional layers, feature maps with multiple channels are obtained as expected. In order to fully use every channel's information, each channel is expected to be attached different weights based on its importance in feature representation. Consequently, channel attention modules can enhance the model's ability. Compared with optical images, SAR images are, moreover, richer in detail and have speckle noise. Through several convolutional and pooling layers, detailed information can be easily lost, and all mix with noise, causing a decrease in accuracy and recall. In order to better recognize useful information and reduce the influence of speckle noise, spatial attention can make the model focus on the position of more important pixels. To sum up, channel attention together with spatial attention can dramatically enhance the model's ability for feature representation in the two dimensions.

Convolutional block attention module (CBAM) is a simple but efficient attention module that is widely used in CNN [37], the structure of which is shown in Fig. 10. In addition to the 1-D channel attention map $M_c \in \mathbb{R}^{C\times 1\times 1}$ obtained by weighted calibration on the channel dimension, the CBAM module also infers the spatial attention map $M_s \in \mathbb{R}^{1\times H\times W}$ along the spatial dimension in two dimensions, and then the two maps are fused with the original input feature maps.

In the channel attention part, CAM is concerned with "what makes sense in an image," and the process is described as

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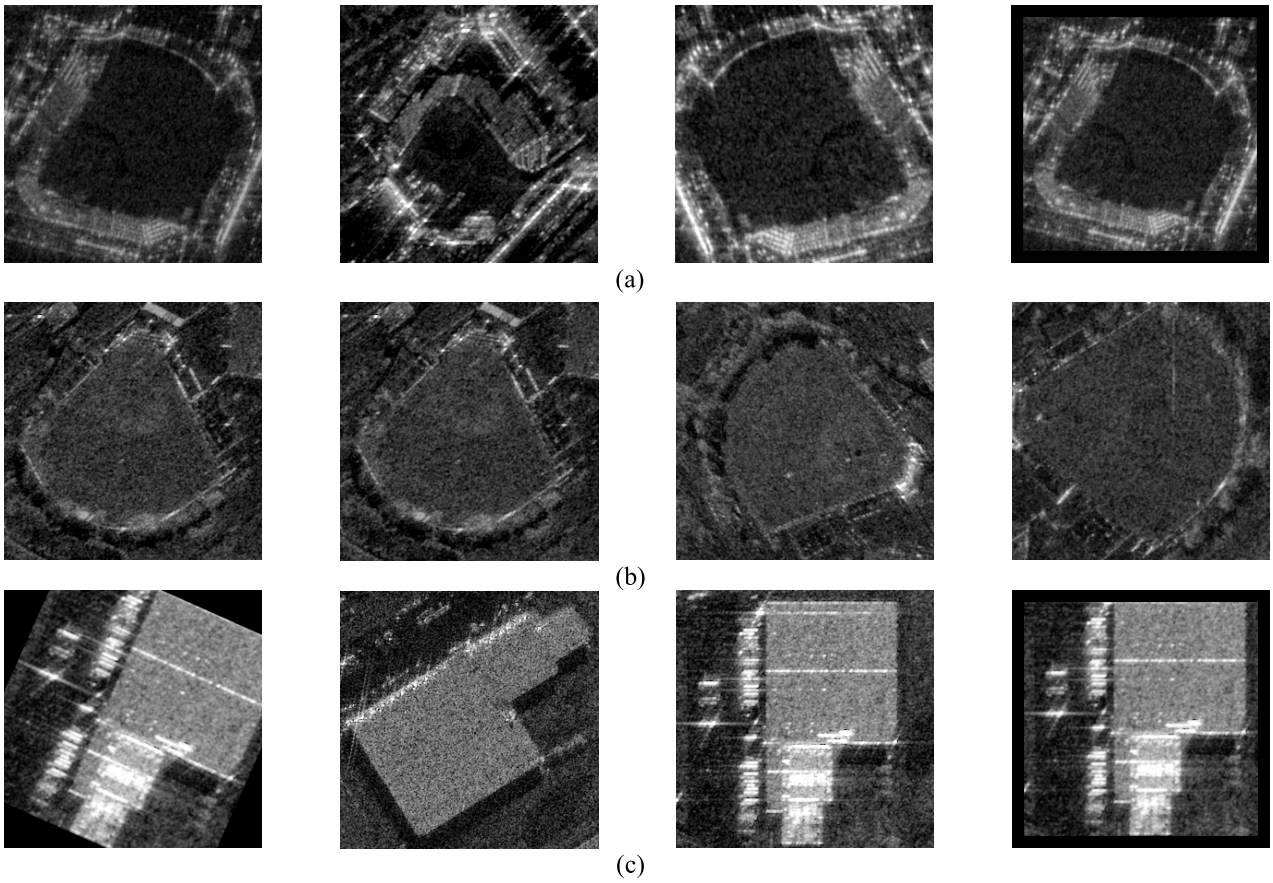


Fig. 6. Examples of the dataset. (a) Example 1. (b) Example 2. (c) Example 3.

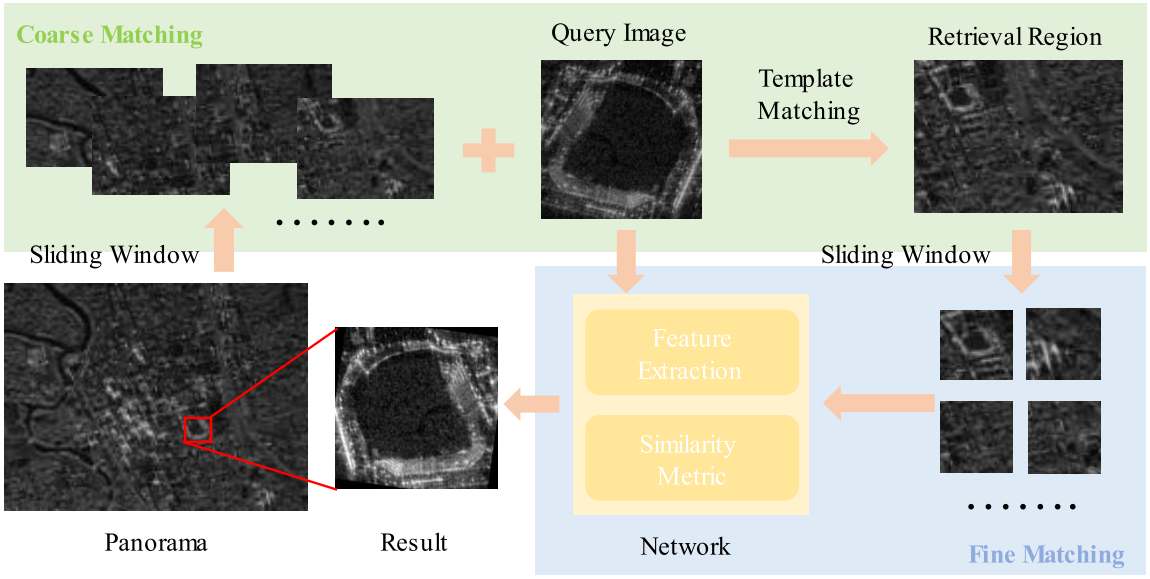


Fig. 7. Framework of our algorithm.

follows:

$$M_c(\text{in}) = \sigma(\text{MLP}(\text{AvgPool}(\text{in}))) + \text{MLP}(\text{MaxPool}(\text{in})) = \sigma\left(f_1\left(f_0\left(\text{in}_{\text{avg}}^c\right)\right) + f_1\left(f_0\left(\text{in}_{\text{max}}^c\right)\right)\right) \quad (7)$$

where σ is the sigmoid function.

CAM compresses the feature map by aggregating the spatial information, obtaining a 1-D vector, which is then passed into the average pooling layer and the max pooling layer, in parallel, in order to compress and aggregate the spatial feature information. After that, two different descriptors are given, which denote the average pooling feature and the max

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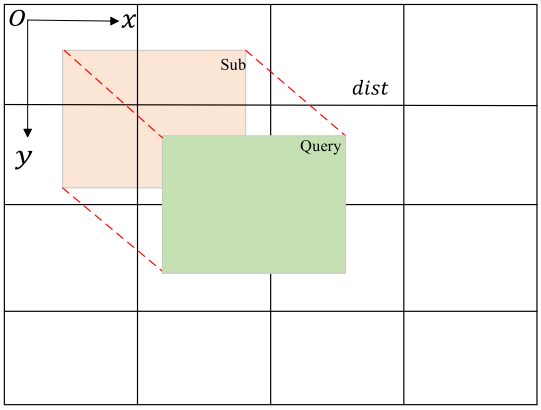


Fig. 8. Process of template matching process.

pooling feature, and contain spatial contextual information. Then, the channel attention maps of the two descriptors are generated through MLP, and finally, the output feature vectors are merged and computed by element-level summation. Since too much information would be lost by single pooling, the CAM module adds a max pooling layer which is connected in parallel with the average pooling layer to increase information richness.

In the spatial attention part, SAM emphasizes "where the image information is." The process is expressed in the following equation:

$$\begin{aligned} M_s(\text{in}) &= \sigma\left(\text{Conv}^{7\times7}\left(\left[\text{AvgPool(in)}; \text{MaxPool(in)}\right]\right)\right) \\ &= \sigma\left(\text{Conv}^{7\times7}\left(\left[\text{in}^s_{\text{avg}}; \text{in}^s_{\text{max}}\right]\right)\right) \end{aligned} \tag{8}$$

where σ denotes sigmoid function, $\text{Conv}^{7\times7}$ is a convolutional operation.

The channel-calibrated feature maps apply the max pooling and average pooling sequentially to compress the channel information and then generate a 2-D descriptor representing the average pooling features of the channel and a 2-D descriptor representing the max pooling features. Ultimately, the two descriptors are connected by a convolutional operation to derive a 2-D spatial attention map.

Channel attention and spatial attention can be combined in two ways: serial or parallel. In the baseline architecture of ResNet, the two submodules are connected in the order of CAM-then-SAM. Since CBAM is a lightweight and plug-and-play module, the model performance can be significantly improved with a small amount of computational and parametric quantities. There are likewise two possible positions of the CBAM incorporated with the ResNet: the module is inserted into the residual block, as shown in Fig. 11(a), and between the layers, as in Fig. 11(b). For Fig. 11(b), we do not know whether it is better to insert CBAM between each of the two layers separately or directly in the last layer to process the global features; therefore, we tested them both to prove the way to improve the best effect.

C. Matching Framework Based on Siamese Network

Because of the high variability of SAR images in looking and azimuth angles, significant differences can arise between images acquired at different moments or locations. Environmental factors and the structural properties of targets can, furthermore, lead to occlusion in parts of the SAR images, causing substantial differences between image pairs that would otherwise be considered similar. These challenges make traditional similarity metrics and matching methods ineffective. An SNN, designed to learn the similarity between two inputs, can address this issue by learning robust feature representations. Through training, the network brings similar image pairs closer together in a metric space while pushing dissimilar pairs further apart, making it more adaptable to the issues of variability and occlusion.

Additionally, SNN is often used as a one-shot classification model consisting of two subnetworks with shared weights. This architecture excels at capturing similarities between inputs while requiring only a small amount of training samples for accurate prediction. The model's robustness to class imbalance makes it especially suitable for situations where SAR data is limited, enhancing its generalizability and effectiveness in addressing the challenges of small datasets.

The modified structure of the image matching is presented in Fig. 12, mainly containing two sections: feature extraction and similarity metric.

The feature extraction part adopts ResNet-50 as the baseline removes the classifier, and adds the CBAM module into the appropriate position to improve the performance of ResNet-50, whose structure has been given in Section III-A. The so-called "subnetwork" with shared weights is two identical networks. It is reflected in the fact that the two inputs are successively passed through the same network without records. The inputs comprised a query image and a match-to-be image from the panorama, respectively. After feature extraction, we can get multidimensional feature maps. The feature maps of each branch are flattened into a 1-D vector to subsequently facilitate computational comparisons. Then, the two 1-D vectors are passed through the metric part.

Since the effect of each similarity metric function is not much different, the simplest L1 norm is chosen. For convenience, the ultimate output of the fully connected layer is changed to 1-D to get a scalar. Then, this vector is activated by the sigmoid function and converted to a scalar value that represents the similarity score between [0, 1].

SNNs typically use either contrastive loss or triplet loss for backpropagation. Triplet loss, which employs three inputs simultaneously for comparative learning, can capture subtle differences more effectively than contrastive loss. Randomly selecting triplets during training can, however, result in an imbalanced dataset, leading to overfitting and slow convergence. This article employs the BCE loss function to tackle these issues. Compared to contrastive loss and triplet loss, BCE loss is less sensitive to the distribution of training and validation samples in complex models, and when combined with the sigmoid activation function, it allows for fast convergence and reduces the risk of overfitting.

IV. Experiments

Based on the above analysis, a scene matching algorithm consisting of two stages is proposed. In the first stage, a sliding

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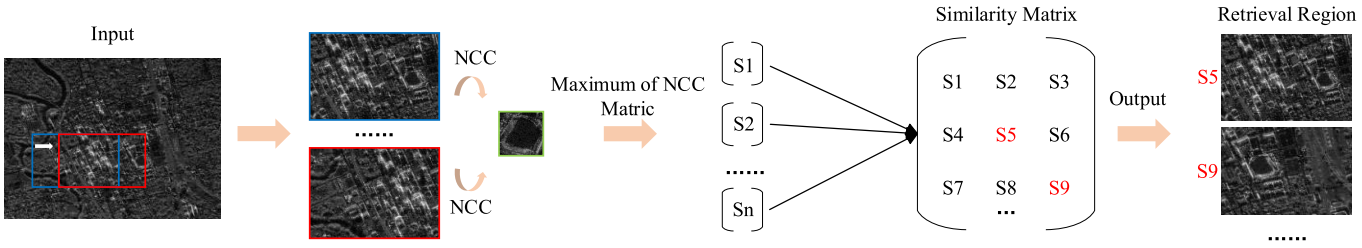


Fig. 9. Process of coarse matching for SAR images.

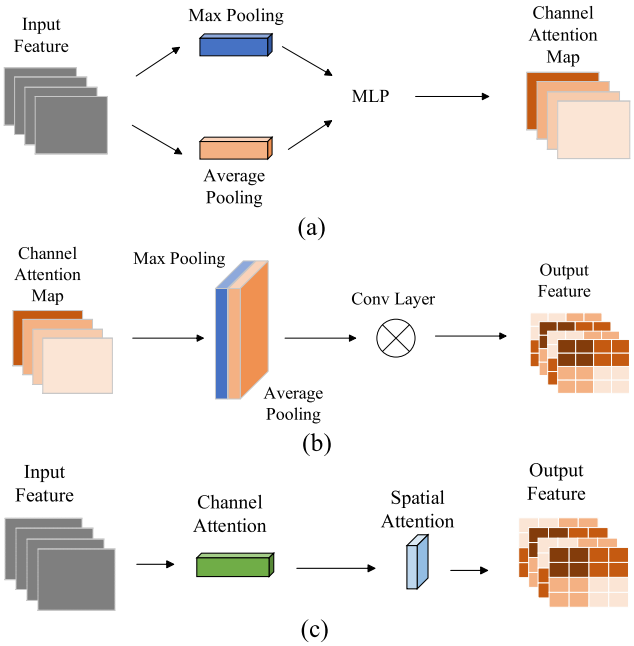


Fig. 10. Structure of CBAM. (a) CAM, (b) SAM, and (c) overall process of CBAM.

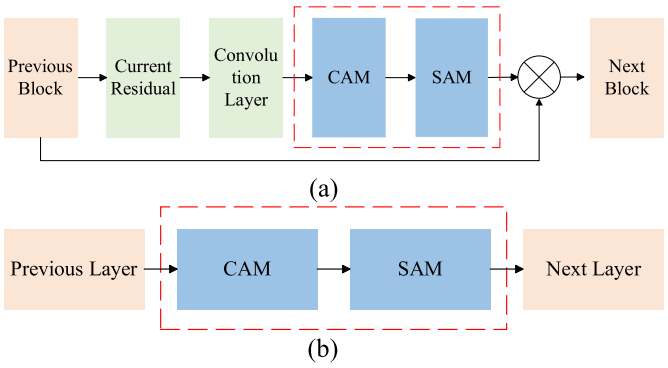


Fig. 11. Possible position of CBAM in ResNet-50. (a) CBAM inserted into the residual block. (b) CBAM inserted between ResNet layers.

window technique is employed on the panoramic image, using NCC to identify candidate regions whose correlation with the query image is above a predefined threshold. In the second stage, a Siamese model performs a sliding window search on the results obtained from the coarse matching and outputs the regions with similarity scores exceeding a set threshold as the final matching results. The specific process is illustrated in the flowchart shown in Fig. 13. We conducted several experiments whose details and analysis are shown to validate the effectiveness of the proposed algorithm.

TABLE II
HARDWARE CONFIGURATION FOR EXPERIMENTS

Hardware	Configuration
GPU	NVIDIA Quadro T1000
Video Memory	4G
CPU	Intel Core i7-9850H
CPU frequency	2.6GHz
Memory	128G

A. Implementation Details

The hardware configuration is shown in Table II. We set the window size for coarse matching to 1200 × 1200, which is large enough to contain most of the subjects, and a stride of 600 in case the target appears at the junction. As for the window size in fine matching, it changes as the query image size changes. Based on experiments, the threshold for adjusting the stride is 0.95 (in Section IV-B).

B. Coarse Matching Experiments

In the real matching process, we input a 3000 × 3000 panoramic image and a query image, sliding the window on the panoramic image with the size of 1200 × 1200, and a stride of 600. Then, compute NCC to get the matrix and record peak points and the corresponding window. After that, select the window areas corresponding to the values that are higher than the threshold as the initial screening results.

Some of the best real matching experiments based on the complete coarse-to-fine process are conducted in Section IV-D. In these experiments, the coarse matching results and fine matching results are all based on optimal values other than the threshold. By analyzing the images in coarse matching, it can be seen that when the query image and the panorama have roughly the same angle and scale, template matching can give a bright spot, and when there is a deviation in the view angle or scale, the bright spot is blurred, and the coverage area becomes larger, but in both cases, the coarse matching method can screen out a region effectively (the region in the red box in the figure). Since the input to the Siamese model is no longer a panorama but a single coarse matching filtered subgraph, the search space is greatly reduced, and the calculation amount and calculation time are reduced.

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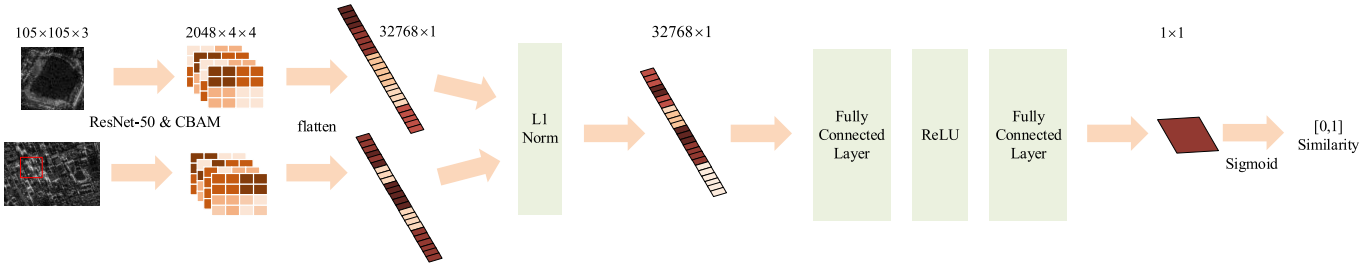


Fig. 12. Structure of the model for fine matching.

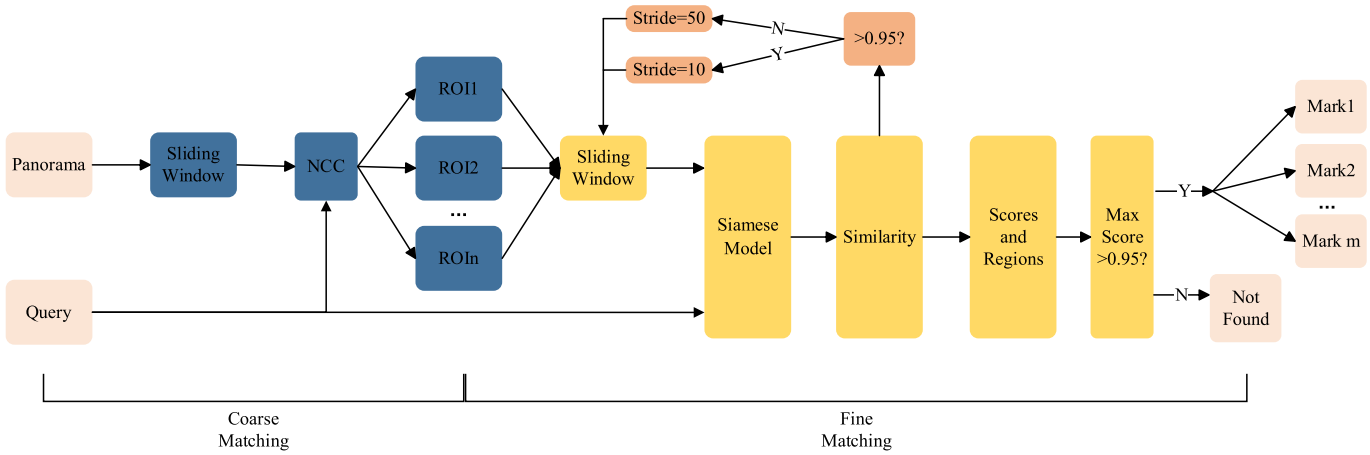


Fig. 13. Process of whole scene matching for SAR images.

In general, due to the output of multiple possible initial regions in the coarse matching, the subsequent fine matching can be searched on several initial regions, which can improve the matching success rate.

C. Experiments on Models

The initial learning rate is set to 0.0001 and batch size is 32. Epoch is 100. Stochastic gradient descent is chosen with a momentum value of 0.9. The learning rate is adjusted with a cosine annealing strategy. We initialize the parameters using a pretrained ResNet-50 model to speed up model convergence.

The training results are presented in Fig. 14, which is evident that compared to inserting CBAM into the network layer, inserting it into the residual blocks increases the complexity of the model and slows down the convergence of the training process, but the loss and accuracy do not show any significant change after sufficient training. We test accuracy, precision, recall, and area under the curve (AUC). All indicators' definitions can be seen in [38].

The results are in Table III. We can get the idea that ResNet-50 with CBAM in residual blocks combined with the Siamese network performs the best, so this model is chosen.

In order to verify the model's capability to detect similar image pairs in different angles, in-samples and out-of-samples are used to test in Fig. 15. Each group of in-sample images is assigned labels 1∼8, arranged left to right and top to bottom, simultaneously input into the model, and the similarity scores are obtained shown in Table IV. By analyzing the data, it can be seen that the similarity scores can reach more than

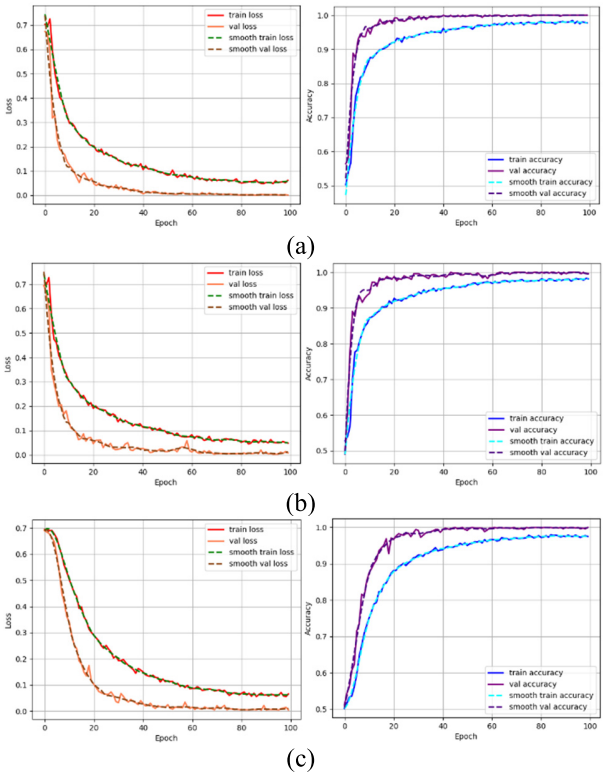


Fig. 14. Training results. (a) CBAM-layer ResNet50+Siamese. (b) CBAM-layers ResNet50+Siamese. (c) CBAM-block ResNet50+Siamese.

0.99 when the background environments do not differ much, and when the environments differ greatly, the similarity scores are also higher than 0.95, so the model is robust in recognizing

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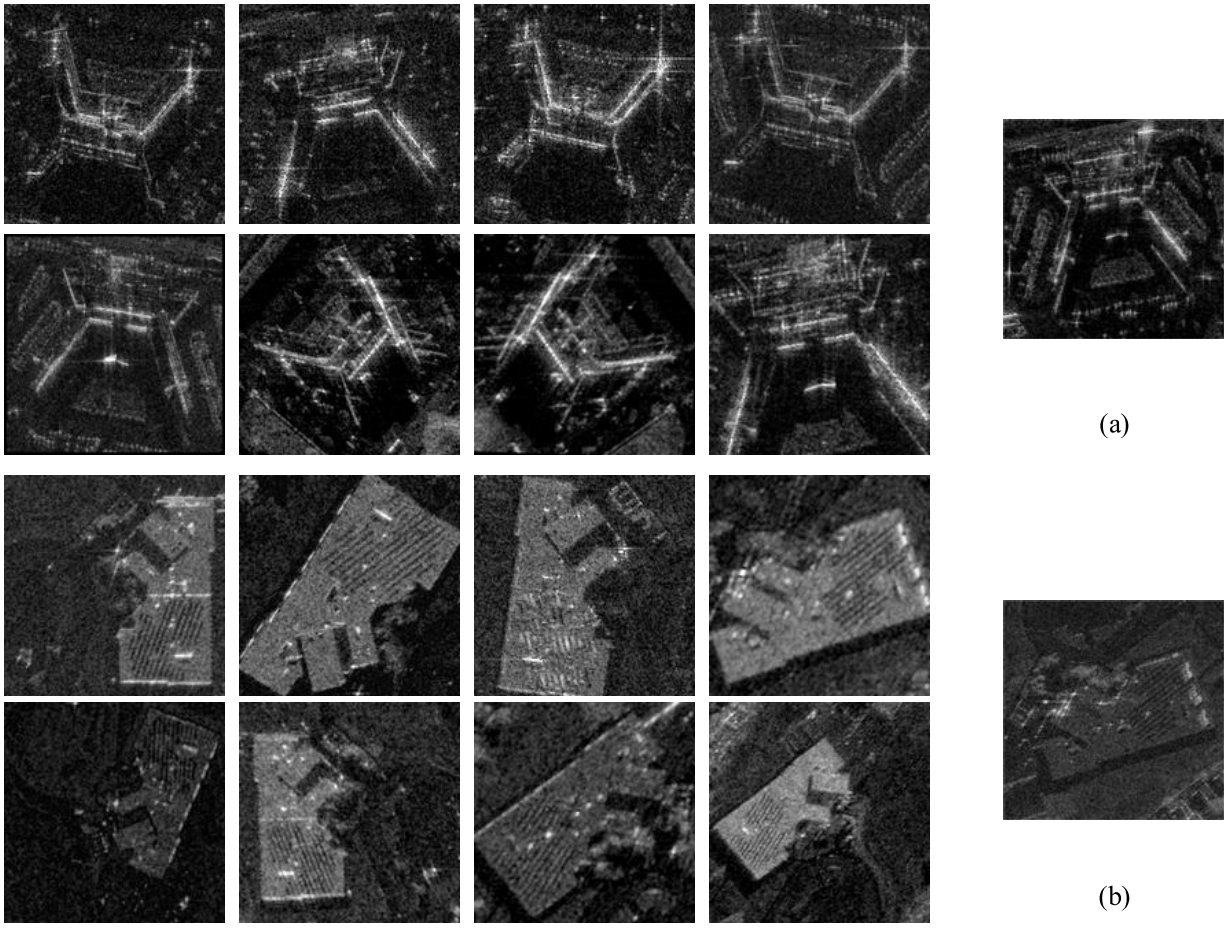


Fig. 15. Training samples and testing samples. (a) Group 1. (b) Group 2.

TABLE III
EXPERIMENTAL RESULTS FOR RESNET WITH DIFFERENT DEPTH AND CBAM IN DIFFERENT POSITIONS

Model (+Siamese)	Accuracy	Precision	Recall	AUC
ResNet-34	74.3	73.0	86.6	81.1
ResNet-50	74.5	72.1	90.9	80.1
ResNet-50 + CBAM-layer	77.2	74.4	91.4	82.7
ResNet-50 + CBAM-layers	77.0	74.2	91.2	83.7
ResNet-50 + CBAM-block	**78.1**	**75.7**	**91.3**	**85.5**
ResNet-101	74.8	71.2	**92.5**	82.3

similar samples and very adaptable to SAR images; therefore, in the following steps the threshold is also set as 0.95.

We carried out several comparative experiments, and ablation experiments were performed on the CBAM and loss function to validate the proposed model. The results are shown in Table V based on 1000 pairs of SAR images. From the analysis of the data in the table, SIFT is less effective than the deep learning model because it is more seriously affected by noise and angles. As to other models, ours takes deeper and more effective features by residual blocks and CBAM compared to AlexNet and SqueezeNet, which have almost indistinguishable performances. Compared to [31], our model is, therefore, more capable of recognizing similar images at different angles by using the same self-made dataset, thereby obtaining higher recall. In ablation experiments, we can know that CBAM makes contributions, and although contrastive loss leads to higher recall, others are barely satisfactory, in addition to their low convergence speed. The BCE loss accelerates convergence and provides good results.

D. Matching Experiments

Three experiments are conducted to validate the whole method's matching effect, and the image information is provided in Table VI. The results in Fig. 16 are based on the best results, where correct matches are identified just by the correlation maximum and the highest similarity scores other than the threshold.

After coarse matching, an accurate position is achieved, with the result considered successful when the window can cover the majority of the target region. Experimental results

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TABLE IV
SIMILARITY OF SAMPLES

Group 1	1	2	3	4	5	6	7	8
Similarity	0.9958	0.9942	0.9990	0.9981	0.9981	0.9992	0.9979	0.9984
Group 2	1	2	3	4	5	6	7	8
Similarity	0.9803	0.9955	0.9686	0.9792	0.9561	0.9942	0.9797	0.9676

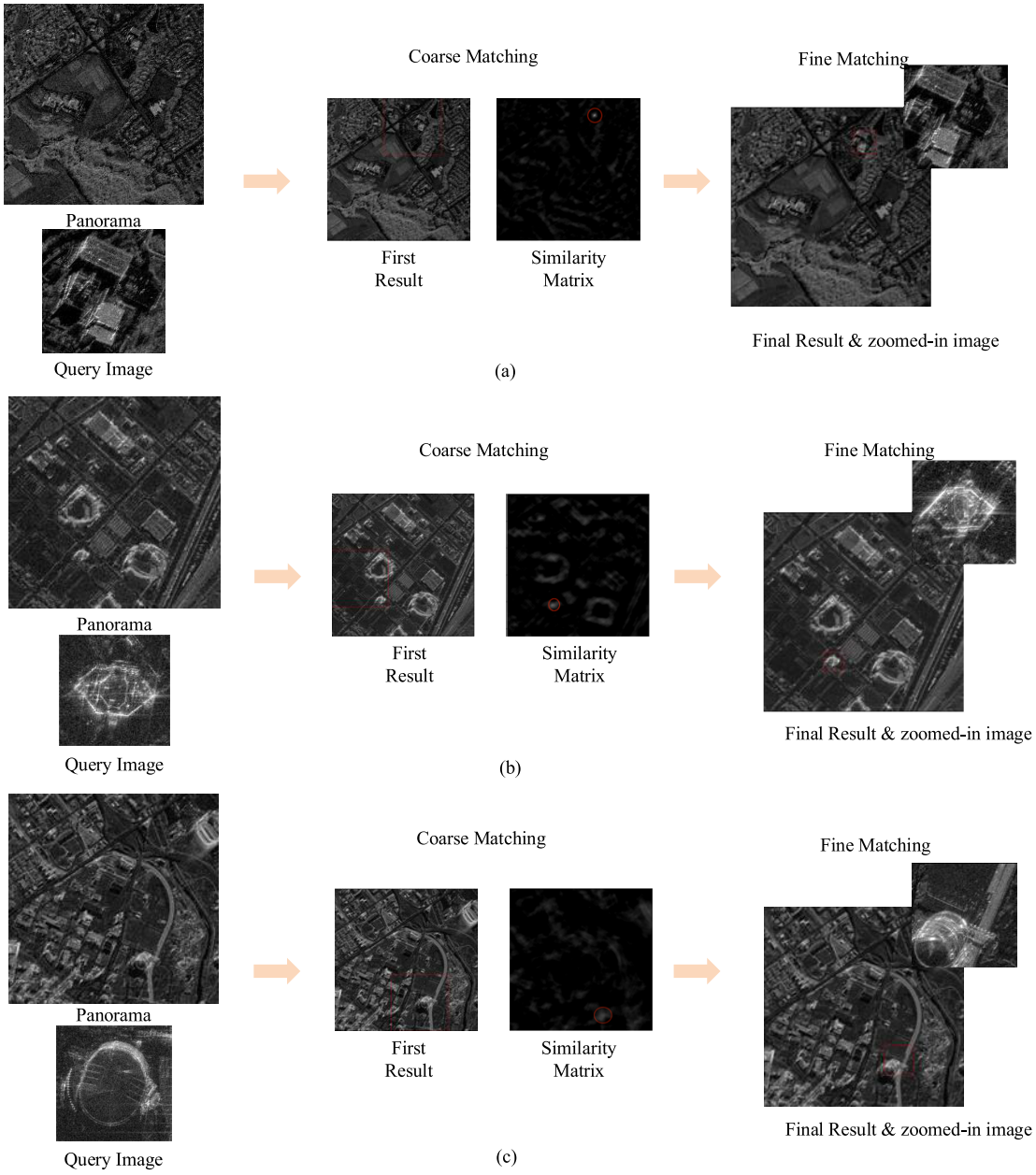


Fig. 16. Experimental results, where inputs are a panorama and a query image; after coarse matching, we get the first result for fine matching and a similarity matrix graph in which the peak is circled in red; through fine matching, we get the final matching result and its corresponding zoomed-in image. (a) Experiment 1. (b) Experiment 2. (c) Experiment 3.

demonstrate that the method remains effective, even when the angle between the matched images is significantly different, as in experiment in Sections IV-B and IV-C.

Following the two-step process, further experiments reveal that the coarse matching employed in this study enhances accuracy. For instance, in Experiment 3, without coarse matching, the method tends to match regions with features similar to the query image, leading to incorrect results. By focusing on the specified region, coarse matching effectively isolates interference and filters out irrelevant regions. This approach improves the process by narrowing down the search areas.

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TABLE V
COMPARATIVE EXPERIMENTS AND ABLATION EXPERIMENTS

Model	CBAM	Contrastive Loss	BCE Loss	Accuracy	Precision	Recall	AUC
SIFT [12]	–	–	–	46.8	48.7	6.0	38.2
Siamese+AlexNet	×	×	√	61.5	82.3	40.4	26.1
Siamese+SqueezeNet [31]	×	×	√	60.5	**83.3**	39.1	24.6
Ours (1)	×	×	√	74.5	72.1	90.9	80.1
Ours (2)	√	√	×	58.0	58.0	**99.3**	34.6
Ours (3)	√	×	√	**78.1**	**75.7**	**91.3**	**85.5**

TABLE VI
IMAGE INFORMATION FOR MATCHING EXPERIMENTS

Index	Panorama	Query Image	Window Size
1	Johnston, Iowa Farm 2023-07-26-16-23 3000×3000	Johnston, Iowa Farm 2023-07-26-16-23 396×397	350×350
2	Philadelphia, PA 2023-05-07-14-40 3000×3000	Philadelphia, PA 2023-05-09-02-39 330×330	300×300
3	Denver, CO 2023-05-09-04:15 3000×3000	Denver, CO 2023-05-09-04-41 500×500	450×450

TABLE VII
TIME REQUIRED FOR EACH METHOD

Experiment	GPU s	GPU+adjustment for stride s	Ours s
1	229.84	219.88	**21.35**
2	225.76	219.10	**31.20**
3	216.25	205.26	**26.11**

Our method, furthermore, significantly reduces the matching time, as illustrated in Table VII. The automatic stride adjustment technique results in a reduction of approximately 10 s in computing time. In comparison, the matching method proposed in this article improves the matching speed by approximately eight times compared to the GPU with the stride adjustment method.

V. CONCLUSION

In this study, a novel scene matching algorithm for SAR images is proposed based on a coarse-to-fine matching strategy. During the coarse matching stage, to reduce computational time and mitigate the influence of irrelevant regions, a grayscale-based template matching method is adopted. A sliding window technique generates a map that compares the query image with the panoramic image, improving the accuracy of candidate region selection. This approach also accounts for the scale differences between the query and panoramic images, overcoming the inherent limitations of traditional template matching techniques. For precise and automatic target region localization, an SNN is employed as the backbone to learn similarities between image pairs. The network is enhanced by integrating ResNet-50 with CBAM to extract more distinctive features, thereby improving overall performance. Compared with conventional matching methods and other backbone networks, the proposed model demonstrates superior accuracy, making it particularly effective in the fine matching stage.

The Umbra Satellite SAR images are used to create image pairs with slices at different angles for model training and testing. The results indicate that the proposed model can enhance the forensic ability with rapid convergence. Then, raw panoramas and slices are presented to the whole framework to test the efficiency and validity of the proposed algorithm. The results illustrate that our method can accurately identify one or more regions similar to the input image while reducing matching time. Additionally, due to coarse matching, we can improve the accuracy of results by excluding certain areas of interference.

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